

4. Let $(\mathbb{R}, d)$ be the usual metric spaces, and $f, f_n : \mathbb{R} \rightarrow \mathbb{R}$ be such that $\{f_n\}$ converges uniformly to f on $\mathbb{R}$. Show that $D(f) \subset \bigcup_{n=1}^{\infty} D(f_n)$, where $D(f)$ is the set of discontinuous points of f . [5]

~~$\{f_n\}$ converges uniformly.~~

$\Rightarrow \forall \epsilon > 0, \exists N_\epsilon > 0$ s.t.

$|f_n(x) - f(x)| < \epsilon$ whenever $n \geq N_\epsilon, \forall x \in \mathbb{R}$.

take $\epsilon > 0$.

$\exists N_\epsilon$ s.t. $|f_n(x) - f(x)| < \epsilon$
 $\forall n \geq N_\epsilon, \forall x \in \mathbb{R}$.

~~take the pt~~ take x_0 where
 f is continuous.

now

to show $D(f) \subset \bigcup_{n=1}^{\infty} D(f_n)$

is equivalent to showing

$$(D(f))^c \supset \bigcap_{n=1}^{\infty} (D(f_n))^c$$

suppose $x_0 \in \bigcap_{n=1}^{\infty} (D(f_n))^c$

that is x is continuous
for f_n is cont. at x
 $\forall n$.

take $\epsilon > 0$.

$\exists N_\epsilon > 0$ s.t. $|f_n(x) - f(x)| < \frac{\epsilon}{3}$
 $\forall n \geq N_\epsilon, \forall x \in \mathbb{R}$

also since f_{N_ϵ} is cont at x_0
 $\exists \delta_{\epsilon, x_0}$

$$|f_{N_\epsilon}(y) - f_{N_\epsilon}(x_0)| < \frac{\epsilon}{3}$$

whenever $d(y, x_0) < \delta$

now $|f(y) - f(x_0)|$ $\hookrightarrow$ normal metric.

$$= |f(y) - f_{N_\epsilon}(y) + f_{N_\epsilon}(y) - f_{N_\epsilon}(x_0) + f_{N_\epsilon}(x_0) - f(x_0)|$$

$$\leq |f(y) - f_{N_\epsilon}(y)| + |f_{N_\epsilon}(y) - f_{N_\epsilon}(x_0)| + |f_{N_\epsilon}(x_0) - f(x_0)|$$

$$< \frac{\epsilon}{3} + \frac{\epsilon}{3} + \frac{\epsilon}{3} = \epsilon$$

$\Rightarrow |f(y) - f(x_0)| < \epsilon$
whenever $|y - x_0| < \delta$
hence f is continuous
at x_0

$\Rightarrow x_0 \in (D(f))^c$
hence proved.

..... Rough Work

at pt x .

$$|f_n(x_m) - f_n(x)|$$

$$|f_n(x_y) - f_n(x)| < \epsilon$$

$\forall$ when $d(y, x) < \delta_{n, \epsilon}$

$$\text{now } |f(y) - f(x)| < |f_n(y) - f_n(x)| + |f_n(y) - f(y)| + |f_n(x) - f(x)|$$

$$|f(y) - f(x)| < \epsilon$$

if it is continuous at $x \rightarrow$

$$|S_n(x) - S_n(x)|$$

$$S_n = \sum_{i=1}^n g_i \geq \sum_{i=1}^n \frac{1}{n^2}$$

$\hookrightarrow$ converges pointwise.

$$|S_n(x) - S_m(x)| < \epsilon \quad \forall n, m \geq N_{\epsilon, x}$$

$$|S_n(x)| \text{ take } B_{\epsilon}(x)$$

$$x \in B_{\epsilon}(x).$$

5. (a) Let $f : [-13, 13] \rightarrow \mathbb{R}$ be defined by

$$f(x) = \begin{cases} x^4 - 13, & x \in \mathbb{Q}, \\ 13x^2 - 49, & x \notin \mathbb{Q}. \end{cases}$$

Justify whether f is Riemann integrable on $[-13, 13]$.

(b) Give short Justifications, using results from the class, to check whether the following functions $f, g : [0, 1] \rightarrow \mathbb{R}$ are Riemann integrable or not:

$$f(x) = \begin{cases} 1, & x = 0, \\ 1 - \frac{1}{n}, & x \in (\frac{1}{n+1}, \frac{1}{n}], \end{cases} \quad \text{and} \quad g(x) = \begin{cases} 1, & x = 0, \\ n - \frac{1}{n}, & x \in (\frac{1}{n+1}, \frac{1}{n}]. \end{cases}$$

(a) $f: [-13, 13] \rightarrow \mathbb{R}$

$$f(x) = \begin{cases} x^4 - 13, & x \in \mathbb{Q} \\ 13x^2 - 49, & x \notin \mathbb{Q}. \end{cases}$$

finding pts of discontinuity.
as both rationals are
irrationals are dense.

for $f(x)$ to be continuous at $x=x_0$

$$\lim_{x \rightarrow x_0} x^4 - 13 = \lim_{x \rightarrow x_0} 13x^2 - 49$$

$$\Rightarrow x_0^4 - 13 = 13x_0^2 - 49$$

this can have at most 4
roots (hence only 4 pts
of continuity)

let y be the set of roots

$$\Rightarrow D(f) = [-13, 13] \setminus y$$

now as y is finite
its measure is 0

$$\text{hence measure of } D(f) = 26 - 0 = 26 \neq 0$$

hence not integrable.

[3 + 1.5 + 1.5]

$$(b) \quad g(x) = \begin{cases} 1, & x = 0 \\ n - \frac{1}{n}, & x \in (\frac{1}{n+1}, \frac{1}{n}] \end{cases}$$

$\neq$ the fn is not Riemann
integrable as it is not
bounded.

$$\lim_{n \rightarrow \infty} n - \frac{1}{n} \rightarrow \infty.$$

and as f^n is unbounded,
it cannot be Riemann
integrable.

$$f(x) = \begin{cases} 1, & x = 0 \\ 1 - \frac{1}{n}, & x \in (\frac{1}{n+1}, \frac{1}{n}] \end{cases}$$

the function is bounded
between 0 and 1.

also the f^n is discontinuous
at pts $x = \frac{1}{n}$ where $n \geq 2$.

hence it has a countable
no of discontinuities.

measure of countable set is zero
hence the f^n is Riemann integrable.

..... Rough Work

